Study

Understand, Connect, Apply

Course paths, review material, and focused notes.

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English study

Vocabulary Notes

Online · PDF

Vocabulary Reader

Open the corrected bilingual notes online or use the printable PDF.

Open vocabulary reader
Direction 01

Computer and Data Technology

From core principles to data processing and implementation.

01 · Java

Object-oriented development

Classes, collections, exceptions, file I/O, and modular design.

02 · Python

Data analysis and ML

Cleaning, features, classification, and evaluation.

03 · Hadoop

Distributed processing

HDFS, MapReduce, Hive, and HiveQL.

04 · Foundations

Core computer science

Data structures, computer organization, operating systems, and networks.

Direction 02

Mathematics and Statistics

Assumptions, objectives, constraints, and limits.

01 · Inference

Multivariate testing

Mean vectors, covariance, and multivariate significance tests.

02 · Classification

Clustering and discrimination

Hierarchical clustering, K-means, discrimination, and standardization.

03 · Reduction

PCA and factor analysis

Variance, loadings, scores, rotation, and interpretation.

04 · Optimization

Operations research

Routing, evolutionary methods, simulated annealing, and tradeoffs.

Direction 03

Special Topics

Audio signal processing

Sampling, encoding, STFT, spectral features, SNR, and quality.

Engineering modeling

Equivalent properties, laminate theory, thermal mismatch, FEM, and sensitivity.

Model evaluation

Training versus testing, description versus inference, and result limits.

Connections

Between Subjects

Statistics explains data, optimization guides choices, and computer systems make methods repeatable.

408 · Exam review

Computer science foundations

408 review · Chinese

211 concept notes across the four exam subjects, with explanations and recent exam references. Calculation practice has a separate page.